

Md. Rifat Mahmud

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PROFESSIONAL SUMMARY

- Computer Science graduate (ULAB, Summer 2026, CGPA 3.55/4.00) working full-time as a Junior AI Engineer at Impala Intech after a four-month AI internship on the same team.
- Builds full-stack web products, Flutter apps, and practical AI tools across NLP, chatbots, and document processing. Strong in Python, TypeScript, and Java, with a foundation in OOP, data structures, and algorithms.

EDUCATION

University of Liberal Arts Bangladesh (ULAB)

Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering (CSE)

June 2022 – Summer 2026

- CGPA: 3.55 / 4.00 (completed)
- Thesis: *Explainable Hybrid Ensemble Diagnosis of Five Lung Conditions from Lung-Segmented Chest X-Rays*. Supervisor: Prof. Muhammad Golam Kibria

GBG University and College

Ghatail, Tangail

Higher Secondary Certificate (HSC)

2018 – 2019

- GPA: 4.00 / 5.00

WORK EXPERIENCE

Impala Intech

Dhaka, Bangladesh

Junior AI Engineer (permanent)

June 2026 – Present

- Moved into a permanent seat after the internship and still in the role
- Develop and deploy AI-powered product work full-time, continuing the NLP, chatbot, and document-processing line started as an intern

Impala Intech

Dhaka, Bangladesh

AI Engineer Intern

February 2026 – May 2026

- Four-month internship on NLP, chatbot development, and document processing
- Completed the internship, then joined the same team as a permanent Junior AI Engineer

TECHNICAL SKILLS

- Programming Languages:** Python, TypeScript, JavaScript, Java, PHP, C, Dart, SQL
- Frameworks & Libraries:** Next.js, NestJS, React, Flutter, FastAPI, Flask, PyTorch, TensorFlow, Keras, LangChain, Prisma, OpenCV
- Databases & Data:** PostgreSQL, MySQL, Redis, ChromaDB, SQL, OOP, Data Structures, Algorithms
- Tools:** Git, GitHub, Docker, LaTeX, Ollama, VS Code
- Soft Skills:** Time Management, Team Player, Adaptability, Leadership, Public Speaking, Critical Thinking, Communication, Conflict Resolution
- Languages:** Bangla (Proficient – writing, speaking, and listening), English (Good listening and basic speaking)

PROJECTS

Explainable Hybrid Ensemble Diagnosis of Five Lung Conditions

ULAB thesis, 2026

- Five-class chest X-ray system (bacterial pneumonia, viral pneumonia, COVID-19, tuberculosis, normal) that segments both lungs, crops the film, and refuses a label when the upload fails the lung check
- Fine-tuned DenseNet201, ResNet50, VGG16, and EfficientNet-B3 on 35,686 lung-cropped images with one shared two-phase schedule
- Soft-averages the four models for the pneumonia classes and COVID-19; normal versus tuberculosis uses EfficientNet-B3, a horizontal flip, and a 0.6 threshold
- 89.23% accuracy and 91.67% macro-F1 on 7,651 held-out images. Grad-CAM++, LIME, and SHAP stay inside the lung mask. Research prototype
- Tech Stack:* Python, PyTorch, OpenCV, PSPNet, CLAHE, Grad-CAM++, LIME, SHAP, Flask

TechSupport Pro — Enterprise IT Help Desk & AI Chatbot

- Role-based portals (User, Helper, Admin) for the IT ticket lifecycle: create, assign, accept/reject, status tracking, and file attachments
- Real-time chat and notifications with Socket.IO and Redis; background jobs with BullMQ
- AI assistant (Ollama, Llama 3.1 8B) for troubleshooting and ticket creation from chat
- NestJS REST API with JWT auth and RBAC; Next.js 14 frontend with Tailwind and Zustand
- *Tech Stack: TypeScript, NestJS, Next.js, Prisma, PostgreSQL, Redis, Socket.IO, BullMQ, Ollama*

Radiant Nutraceuticals Pharma Assistant — AI Chatbot

- Bilingual English/Bangla chatbot for 33 herbal products, covering dosage, side effects, warnings, and interactions
- English replies from a local LLM; Bangla replies from curated fields so product facts stay stable
- Admin pipeline extracts PDF, PPTX, DOCX, and TXT to refresh the medicine database; RAG over the product catalog
- *Tech Stack: Python, FastAPI, Ollama, ChromaDB, RAG, sentence-transformers*

YNotes — Academic Notes Marketplace

- Full-stack EdTech marketplace with three roles (Student, Creator, Admin) on Next.js 14 and TypeScript
- Normalized PostgreSQL schema and 71+ REST endpoints with Prisma and NextAuth v5
- SSLCommerz wallet top-ups (bKash, Nagad, Rocket) with IPN callbacks
- Secure PDF delivery via Cloudflare R2, 70/30 creator revenue share, analytics, and moderation
- *Tech Stack: Next.js 14, TypeScript, PostgreSQL, Prisma, NextAuth v5, Tailwind, Cloudflare R2, SSLCommerz, Redis*

Easy Prescription

- Scans a handwritten prescription, matches the medicines, then saves, speaks, or downloads the result as a PDF
- OCR.space handwriting engine with a fallback engine; brand names matched with Levenshtein distance and Soundex, plus a manual pick when the guess is weak
- English and Bangla screens, OTP login, history, PDF download, and ElevenLabs speech. Same flow on the web and in a Flutter app
- *Tech Stack: React, TypeScript, Express, MySQL, Flutter, OCR.space, ElevenLabs*

Also: FIFA World Cup Prediction App; Brand Lifecycle Task Manager; Novara (novaraonline.xyz); plus mobile apps and smaller builds. Full write-ups: portfolio.

CO-CURRICULAR & EXTRACURRICULAR ACTIVITIES

Participant — UCPC Take-Off Programming Contest

Fall 2023

- Competed in a university-wide competitive programming contest and solved 4 out of 5 problems

REFERENCES

Prof. Muhammad Golam Kibria, PhD, SMIEEE

Professor & Head, Department of Computer Science and Engineering (CSE)

University of Liberal Arts Bangladesh (ULAB)

Email: golam.kibria@ulab.edu.bd

DECLARATION

I hereby declare that all the information provided in this CV is true and correct to the best of my knowledge.

Md. Rifat Mahmud